

UJJWAL RAGHAV

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EDUCATION

Birla Institute of Technology & Science (BITS), Pilani & University at Buffalo

2024 – 2028

Collaborative dual degree program · B.E. Computer Science (BITS) & B.S. Computer Engineering (UB)

- Relevant coursework: Object-Oriented Programming, Discrete Structures, Data Structures & Algorithms, Deep Learning, Logic in Computer Science.

Delhi Public School, Vasant Kunj

2020 – 2024

Class XII, CBSE — 88.4%

EXPERIENCE

Research Assistant – Generation of Communication topologies for LLM based agentic systems

April 2026 – Present

Computer Vision and Machine Learning Group · University at Buffalo

- Developing a generative framework for dynamic, task-adaptive communication topology design in LLM-based multi-agent systems, optimizing jointly for task performance, token efficiency and robustness

Research Assistant — Geometry aware Deepfake Generation via 3D Disentanglement

February 2026 – Present

Course Research Project under Prof. Pratik Narang · BITS Pilani

- Designing a latent diffusion-based architecture (U-Net backbone) with a transformer encoder prior to extract source facial features, injected via cross-attention into a frozen diffusion backbone — enabling appearance transfer without retraining the generative model.

AI/ML Vertical Member — BITS Pilani Coding Club

April 2025 – Present

- Architected a Retrieval-Augmented Generation (RAG) system over institutional documents, improving information accessibility for 3,000+ students.
- Applied NLP techniques to build an intelligent query interface capable of synthesizing information across diverse, unstructured document formats.

RESEARCH & TECHNICAL PROJECTS

Baby Dragon Hatchling (BDH) Narrative Consistency Model

- Engineered an inference and interpretability pipeline to detect and visualize narrative contradictions between structured character backstories and unstructured novel text.

Dual-Architecture RAG Advisory System

- Built a comparative RAG framework for academic advising, implementing both a LangChain/ChromaDB pipeline and a custom tensor-based retrieval engine against institutional datasets.

Autonomous Deep-Q Learning Agent

- Developed a Deep Q-Learning agent for the Snake game environment, designing custom reward functions and training environments to optimize convergence and agent performance.

Underwater Audio Classification via CNNs

- Built an end-to-end CNN-based classification pipeline on a self-collected 10 GB+ dataset of underwater audio files, using mel spectrogram representations as input features.

TECHNICAL SKILLS

Languages: Python, C++, Java, C

ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn, TransformerLens

Tools: Git, Linux (Ubuntu), Jupyter, LangChain, ChromaDB